Astronomy 98 Final Project

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1 Background

Intensity is a measure of the amount of light produced per unit area. For galaxies, this value decreases exponentially due to the decrease in stellar density as you get further from the galactic bulge. Although all galaxies follow an exponential light profile, the slope of the exponential decay varies for different galaxy types. One way to model this is through the Sérsic profile given by

$$\mu(r) = \mu_e + \frac{2.5 \cdot b_n}{ln(10)}[(\frac{r}{r_e})^{1/n} - 1]. \tag{1}$$

$$b_n \approx 2n - 0.324 \tag{2}$$

Where $\mu(r)$ is the intensity of the galaxy at any given radial distance in units of magnitudes, μ_e is the half-light intensity (or half the total intensity of the galaxy), b_n defines the shape of the light profile, r is the radial distance from the center of the galaxy, r_e is the effective radius that encloses half of the total light of the galaxy, and n is the Sérsic index. Common values for the Sérsic index are $n = 1$ for spiral galaxies and $n = 4$ for elliptical galaxies.

2 Tools/Resources

For this project, I used several Python packages to execute what I wanted to do. These included:

- Numpy - Used to create arrays and mathematical tools for proper data analysis
- Pandas - Used to organize and clean the initial data that I downloaded
- Matplotlib - Used for all of the plotting for this project
- Scipy - Used for the regression analysis used to find the Sérsic index of the galaxies

On top of just these packages, I also downloaded data from the Strasburg astronomical data center. Specifically I downloaded data from this [2] dataset.

3 Methods

3.1 Cleaning

The raw data I downloaded had several issues that I had to clean before I could do meaningful analysis with it. The first of these issues was the inclusion of empty cells in some of the data entries, so I first removed those blank entries. Then I had to separate the data into each of the galaxies to make the data more usable.

On top of that the data itself was organized in a way that was not usable to perform the fits that I wanted, as the data was organized in one big data frame with sections for each galaxy. To fix this, I created a function that took in the data frame and created a list containing lists of each galaxy and the relevant information for the fit.

After cleaning the data I was able to plot each of the galaxies on the same axis as displayed by fig 1.

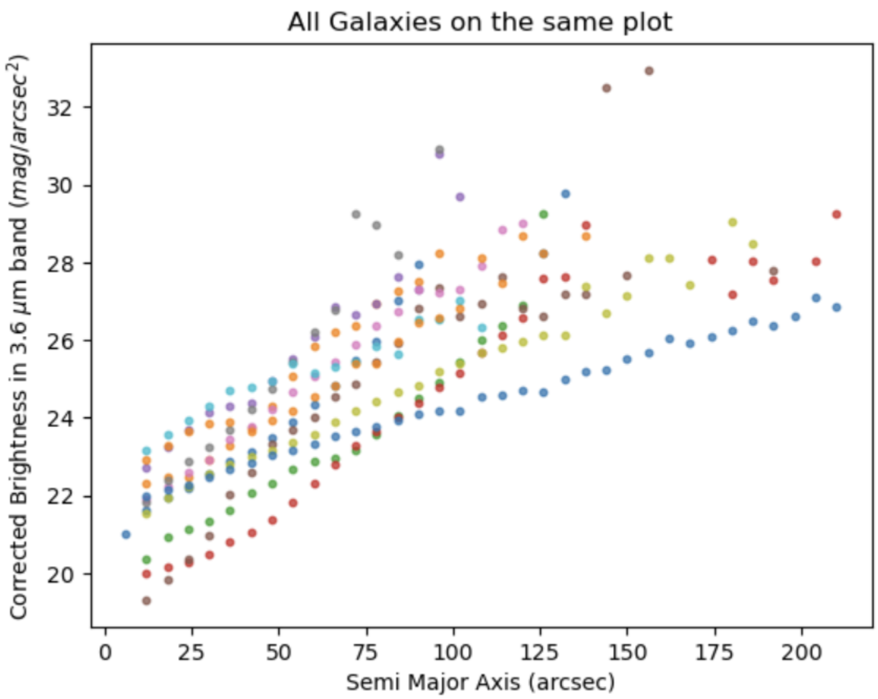

Figure 1: All galaxy plots on same axis

3.2 Fitting/Plotting

After cleaning the data, I then looped through all 48 galaxies and performed the fit to Eq (1) getting all relevant parameters, and then plotting the raw data and the fit for each galaxy on a massive subplot.

4 Results

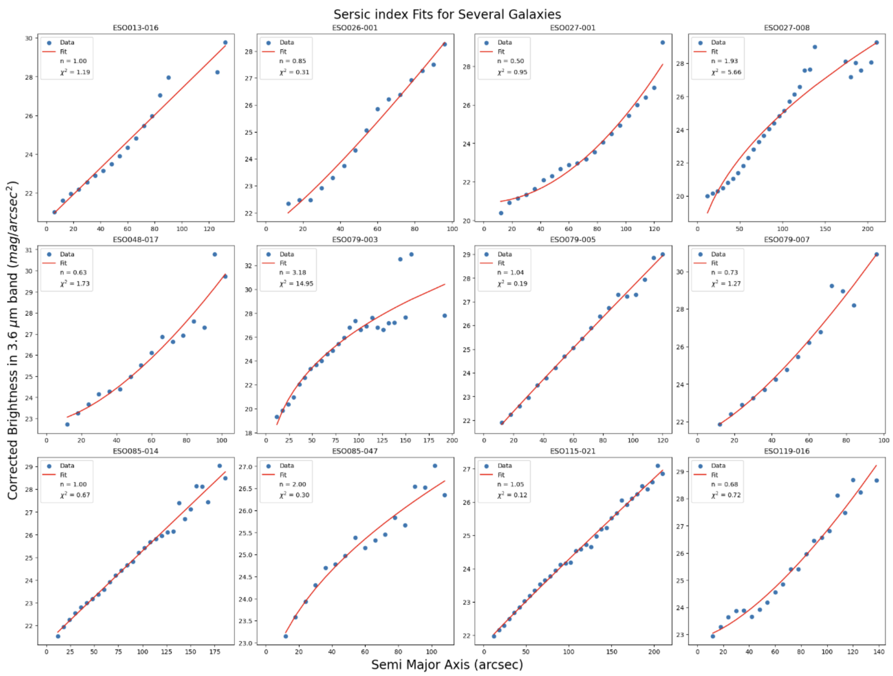

Figure 2: Plots of each galaxy and their corresponding fits

From fig 2 it is possible to see all of the relevant data for the 12 chosen galaxies. I chose to only report the Sersic index for each of the galaxies, as that was the most important parameter for this project.

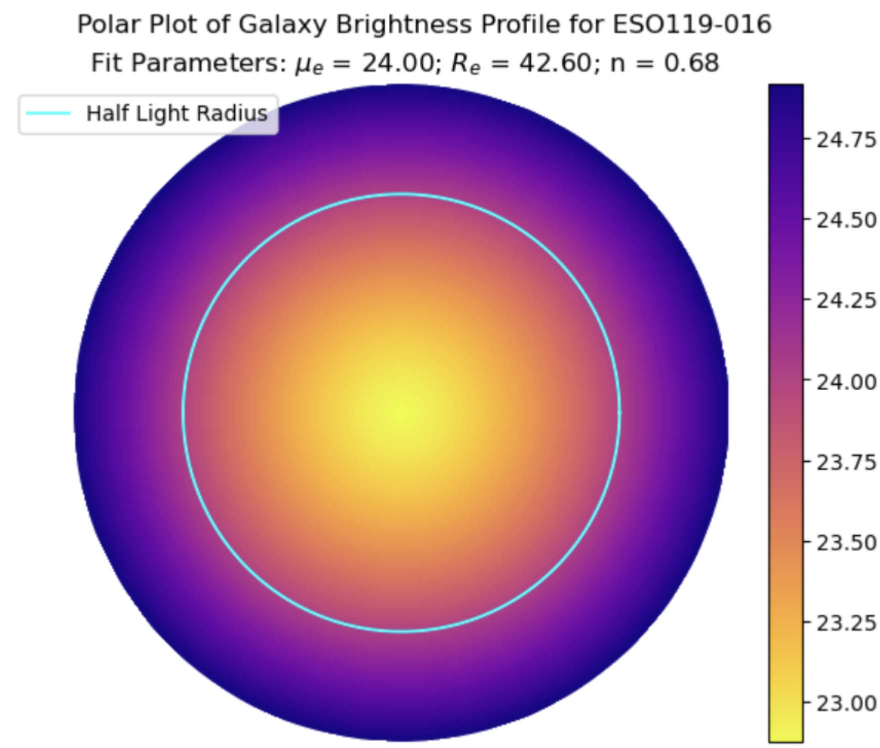

Figure 3: Mock Galaxy profile of chosen galaxy with Half light Radius

Fig 3 represents a mock galaxy brightness profile using the last galaxy on the main plot. This plot shows what the model looks like for an actual brightness as well as the half-light radius.

5 Conclusion

Overall, this project was a major success, as I was able to find the Sersic index for the 12 chosen galaxies. On top of that, I was able to visually represent what the model looks like on a circular galaxy. Unfortunately, not everything was perfect with the project. fig 2 shows the chi-squared for each of the fits, and for most of them the chi-squared is not 1 where it ranges from less than 1 to greater than 1. This implies that the fit for each galaxy is not totally accurate. This inaccuracy can be mainly pinpointed to the approximation of b_n that was made to simplify the fit.

References

[1] Graham, A. (2005) *A CONCISE REFERENCE TO (PROJECTED) SÉRSIC R1/n QUANTITIES, INCLUDING CONCENTRATION, PROFILE SLOPES, PETROSIAN INDICES, AND KRON MAGNITUDES.* https://ned.ipac.caltech.edu/level5/March05/Graham/Graham2.html#note2

[2] Bouquin, A. (2018) *The GALEX/S^{4G} surface brightness and color profiles catalog. I. Surface photometry and color gradients of galaxies.* https://cdsarc.cds.unistra.fr/viz-bin/cat/J/ApJS/234/18